NORTHEASTERN UNIVERSITY LONDON

MSc Artificial Intelligence and Computer Science

Comparative Evaluation of Explainable AI Methods for Customer Churn Prediction in Telecommunications:

A SHAP- and LIME-Based Framework with Marketer-Oriented Segmentation

MSc Dissertation Project Proposal (AE1)

Student name:	Rajarajachozhan Vivekanandhan Kothainayagi
Student number:	03189049
Supervisor:	Ibukun Afolabi
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Table of Contents

1. Introduction

1.1 Context

Customer churn is among the most financially damaging challenges facing the global telecommunications industry. Annual churn rates in the sector range between 20% and 40%, with the cost of acquiring a new subscriber estimated at five to seven times the cost of retaining an existing one (Verbeke et al., 2012). This economic asymmetry has driven sustained investment in predictive modelling, where machine learning classifiers particularly ensemble methods such as XGBoost and Random Forest now routinely achieve AUC scores above 0.85 on standard benchmark datasets (El Attar and El-Hajj, 2026; Imani et al., 2025). These models can identify which customers are likely to leave with considerable accuracy, enabling operators to prioritise retention resources more efficiently than traditional rule-based approaches. Yet a marketing team presented with a ranked list of at-risk customers but no explanation of the drivers behind each prediction faces a decision problem, not a decision support tool. Accurate prediction, on its own, is no longer enough.

1.2 The Challenge

The core challenge is that the models which perform best are also the least interpretable. Gradient-boosted ensembles and deep learning classifiers operate as "black boxes," producing predictions without human-readable justification for why a particular customer was flagged as a likely churner (Adadi and Berrada, 2018). This opacity creates two distinct problems. First, it undermines practical utility: marketers cannot design a targeted retention intervention, a loyalty discount, a contract renegotiation, a proactive service call without understanding what drives the churn risk for a given customer (Rudin, 2019). Second, it creates growing regulatory exposure. The GDPR Article 22 establishes a right to meaningful information about the logic behind automated decisions that significantly affect individuals, and the EU AI Act (2024) introduces further transparency obligations for AI systems deployed in consumer-facing contexts (Wachter, Mittelstadt and Russell, 2018). Telecom churn models that inform pricing, outreach, or service decisions fall squarely within this regulatory scope.

1.3 Aims and Preliminary Thesis

This dissertation pursues three specific aims:

1. **Benchmark three churn prediction models** - Logistic Regression, Random Forest, and XGBoost on the Cell2Cell telecom dataset, establishing an accuracy-to-explainability spectrum from an inherently interpretable baseline (Logistic Regression) through to a high-performing but opaque ensemble (XGBoost).
2. **Systematically compare SHAP and LIME** as post-hoc explanation methods applied to the best-performing model, evaluating their outputs across four metrics: fidelity (how accurately the explanation reflects true model behaviour), stability (whether similar inputs produce similar explanations), sparsity (whether explanations highlight a concise set of actionable features), and agreement (whether the two methods identify the same features as important for the same prediction).
3. **Derive marketer-actionable insight** by clustering churners based on their SHAP feature-value vectors using k-means, producing behaviourally distinct churn segments mapped directly to differentiated retention strategies.

The preliminary thesis is that SHAP will demonstrate superior global consistency, stability, and segmentation utility due to its game-theoretic foundation, while LIME will offer stronger local fidelity for individual customer explanations. Neither method alone will prove sufficient for a production-ready marketer decision support framework motivating the hybrid, segmentation-enhanced approach this project develops.

1.4 Proposed Methodology Overview

The project follows a quantitative, experimental design built on the Cell2Cell dataset, a publicly available telecom churn dataset from Duke University comprising 71,047 customer records across 58 attributes. The dataset will be split 70/30 with SMOTE applied exclusively to the training partition. Three classifiers will be trained and evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. SHAP and LIME will be applied to the best-performing model and compared across four evaluation metrics. A SHAP-based k-means clustering step will segment churners into distinct reason-profiles identifying why each group is leaving (price sensitivity, service dissatisfaction, competitor appeal, or feature gaps) and mapping each profile to a tailored retention offer. Finally, an interactive Streamlit dashboard will let marketers explore individual predictions, compare SHAP and LIME explanations side-by-side, drill into cluster profiles, and receive automated retention recommendations in real time. Section 3 details each stage fully.

1.5 Why This Project, How It Improves on Prior Work

Existing studies in telecom churn prediction have typically applied XAI methods in isolation. El Attar and El-Hajj (2026) applied SHAP to an XGBoost model with the IBM Telco dataset, however they did not compare SHAP with another approach, so the reliability of those ranks was not validated. Krishna et al. (2022) show that common explanation approaches often contradict on feature rankings for the same forecast, casting substantial doubt on whose explanations marketers should trust. This project improves prior work in three ways. First, it introduces a head-to-head SHAP–LIME comparison using a structured four-metric evaluation framework. Second, it extends explanation outputs into a segmentation layer translating feature-level insights into marketer-actionable retention strategies. Third, it operationalises the framework through an interactive Streamlit tool. Section 2.6 expands on these contributions.

2. Literature Review

2.1 Customer Churn Prediction in Telecommunications

The study of customer churn in telecommunications has passed through three distinct phases. The earliest approaches relied on statistical techniques - logistic regression and survival analysis, which offered interpretability but struggled with high dimensionality and non-linear feature interactions (Verbeke et al., 2012). The second wave brought in classical machine learning classifiers (e.g., decision trees, support vector machines) that increased accuracy but were still limited in their capacity to capture complex customer behaviours. The third and present phase is characterised by the ensemble approaches.

Today, our toolkit consists of XGBoost, Random Forest, and gradient-boosted decision trees, which constantly exceed previous techniques on AUC-ROC, precision, and recall on several telecom benchmarks (El Attar and El-Hajj, 2026; Imani et al., 2025). But this progress has come at a cost. The models grew more powerful, but also more opaque. Nurhidayat and Anggraini (2025) found that the XGBoost was the highest performing classifier among six classifiers on a large telecom dataset, however it was unable to provide any natural reasons for its predictions. Pandey and Pandey (2026) confirm this broader pattern: across 85 studies published between 2015 and 2025, fewer than 12% incorporated any form of post-hoc explanation or interpretability analysis.

The field no longer asks, "can we predict churn?" that question is effectively closed, with ensemble classifiers reliably achieving AUC scores above 0.85. The open question is whether churn can be predicted profitably and explainably, enabling marketing teams to act on model outputs with confidence and regulatory compliance. This dissertation engages directly with that open question.

2.2 Explainable AI: Foundations and Methods

Explainable AI encompasses a broad family of techniques designed to make model outputs understandable to human users. There is a fundamental difference between intrinsic interpretability and post-hoc explanation (Adadi and Berrada, 2018). Intrinsically interpretable models such as logistic regression and shallow decision trees are transparent by design; their internal logic can be directly inspected. Post-hoc methods, by contrast, are applied after training to explain an already-fitted model, making them compatible with high-performing but opaque classifiers. Molnar (2025) argues that post-hoc methods are increasingly dominant in applied settings where predictive performance cannot be sacrificed for transparency.

Two methods have achieved particular prominence. SHAP (Lundberg and Lee, 2017) draws on Shapley values from cooperative game theory to assign each feature a contribution score that is locally accurate, consistent, and additive. LIME (Ribeiro, Singh and Guestrin, 2016) takes a fundamentally different approach: it perturbs the input around a single instance and fits a locally weighted linear model to approximate the classifier's local decision boundary. Both are model-agnostic and produce feature-level explanations, but their theoretical foundations diverge sharply. SHAP guarantees mathematical properties that LIME does not, while LIME offers computational simplicity that SHAP's exact kernel variant cannot match. Wachter, Mittelstadt and Russell (2018) further expanded the landscape by proposing counterfactual explanations though this method remains less widely adopted in churn prediction specifically.

2.3 The Disagreement Problem in Explainable Machine Learning

If SHAP and LIME both claim to explain the same model, a reasonable expectation is that they should broadly agree on which features matter most. In practice, they frequently do not. Krishna et al. (2022) used seven prominent explanation approaches for the same models and datasets and observed that pairwise agreement on the top-ranked features is usually below 50% and extremely unpredictable across datasets, model types and individual cases. The “explanation risk” of employing one method is seldom acknowledged or measured by practitioners.

The problem is not evenly distributed across methods. Slack et al. (2020) demonstrated that LIME is particularly vulnerable to instability: repeated runs with different random seeds produce different feature rankings because perturbation-based sampling introduces stochastic variation. Zhang et al. (2019) reinforced this, showing local surrogates are sensitive to neighbourhood definition and sample size. SHAP benefits from its axiomatic guarantees symmetry, efficiency, and additivity, which constrain possible explanations and reduce variance.

Despite this evidence, Doshi-Velez and Kim (2017) noted that the field still lacks a standardised quantitative framework for evaluating and comparing explanation quality. Most comparisons rely on visual inspection of feature importance plots rather than structured metrics. This gap is precisely what this dissertation seeks to fill.

2.4 Explainability Applied to Telecom Churn

A growing number of studies have applied XAI methods to telecom churn prediction, but with varying rigour. El Attar and El-Hajj (2026) applied SHAP to an XGBoost model trained on the IBM Telco dataset, where contract type, tenure and monthly costs were the key churn variables, but there was no comparison with alternative approaches. Al Mamun et al. (2026) similarly used a single method, i.e., SHAP, without quantitative measurement of the quality of the explanation.

Other recent work has applied broader toolkits but without comparative depth. Bolla and Narsepalle (2026) applied SHAP to a telecom ensemble classifier, contributing useful domain findings but without comparing against an alternative method. The IIETA study (2024) followed the same pattern, using SHAP to identify feature importance in a telecom churn context but stopping short of any structured evaluation of explanation quality. Where studies do incorporate more than one method, they tend to present SHAP and LIME outputs as parallel visualisations rather than subjecting them to a head-to-head quantitative comparison. Pandey and Pandey (2026) confirm this pattern in their systematic review: the dominant approach in the literature is to apply a single XAI method as a post-hoc add-on, not to evaluate or compare explanation methods as research objects in their own right.

Three clear gaps emerge. First, almost no telecom churn study compares SHAP and LIME on the same model using the same dataset. Second, quantitative evaluation using defined metrics such as fidelity, stability, sparsity, and agreement is virtually absent. Third, no published study uses SHAP feature-value vectors as inputs to a customer segmentation process.

2.5 Ethics, Fairness, and Deployment of Explainable Churn Models

Three deployment concerns are particularly relevant. First, discriminatory impact: churn models trained on historical data may encode biases related to geography or income proxies, and retention strategies built on biased explanations risk reinforcing inequalities (Selbst and Barocas, 2018). Second, manipulability: Bhatt et al. (2020) found that explanation methods can be gamed to produce favourable-looking explanations while preserving a discriminatory decision boundary underneath. Third, the right to explanation: Wachter, Mittelstadt and Russell (2018) argued that GDPR Article 22 creates a qualified right to meaningful information about automated decisions, but the legal threshold for "meaningful" remains unsettled. These concerns shape the ethical boundaries within which any practical framework must operate.

2.6 Synthesis: Why This Project?

Past work has applied SHAP to telecom churn models as a visualisation layer, and a handful of studies have placed LIME alongside it, but none have compared the two quantitatively using defined evaluation metrics on a large telecom dataset, and none have extended SHAP outputs into a customer segmentation framework. This project fills that gap. It replaces visual intuition with metric-driven rigour across fidelity, stability, sparsity, and agreement, and translates individual feature explanations into marketer-actionable segments through SHAP-vector clustering. Future enhancements include multi-class churn-type prediction, counterfactual integration, real-time deployment, and fairness auditing.

3. Research Methodology

3.1 Research Philosophy and Approach

This project is positioned within a positivist research philosophy, following Saunders, Lewis and Thornhill (2019). Positivism is appropriate because the research seeks to measure observable, quantifiable properties of explanation methods not to interpret subjective meaning. The approach is deductive: hypotheses about the relative performance of SHAP and LIME are derived from literature and tested experimentally. The strategy is computational experimental, with all variables manipulated programmatically on a fixed dataset. The time horizon is cross-sectional, and the techniques span supervised classification, post-hoc explanation generation, and unsupervised clustering in a sequential pipeline. The project is feasible within the sixteen-week timeframe: all tools are open-source, the dataset is publicly available, and computational requirements fall within laptop capacity with Google Colab as a fallback.

3.2 Dataset

The project uses the Cell2Cell Telecom Churn Dataset, originally compiled by the Duke University Teradata Center and distributed publicly via Kaggle. The dataset comprises of 71,047 customer records described by 58 attributes covering demographics, account tenure, usage patterns, service plans, service area, customer service interactions, handsets, personal information and billing. The target variable is binary churn, with approximately 28.6% of customers labelled as churners.

Cell2Cell was selected for four reasons. First, its volume substantially exceeds MSc-level expectations and provides sufficient statistical power for both classification and clustering. Second, it contains real, anonymised telecom data rather than synthetically generated records. Third, it is an established academic benchmark enabling comparison with prior results. Fourth, its 58-attribute feature space is considerably richer than IBM Telco (21 attributes, 7,043 records), allowing SHAP and LIME to operate on a meaningfully complex feature landscape.

Preprocessing includes median imputation for numerical features and mode for categorical, one-hot encoding, min-max scaling, and a 70/30 train-test split with SMOTE applied exclusively to the training set.

3.3 Predictive Models

3.3.1 Logistic Regression

Logistic regression predicts the probability of churn as a log-odds function, i.e., a linear sum of input features weighted and passed through a sigmoid activation. Each coefficient is directly related to the direction and magnitude of the effect of a feature. This is the interpretability baseline and is expected to achieve AUC-ROC around 0.78-0.82.

3.3.2 Random Forest

Random Forest builds an ensemble of decision trees, each trained on a bootstrapped sample and a random subset of features, and aggregates them by majority voting (Breiman, 2001). Robust to outliers and insensitive to feature scaling, it occupies the accuracy-interpretability mid-point, with expected AUC-ROC of approximately 0.85–0.88.

3.3.3 XGBoost

XGBoost is a gradient-boosted tree ensemble where each successive tree corrects the residual errors of its predecessors (Chen and Guestrin, 2016). This produces consistently high predictive performance but renders the model opaque. XGBoost is the primary object of explanation in this project, with expected AUC-ROC of approximately 0.88–0.92.

3.3.4 Hyperparameter Tuning, Validation, and Evaluation Metrics

All three models will be tuned using grid search with 5-fold stratified cross-validation on the training partition. Random seeds fixed at 42 throughout. Performance evaluated on the held-out test set using accuracy, precision, recall, F1-score, AUC-ROC, calibration plots, and confusion matrices, following Imani et al. (2025). Internal validity is maintained through fixed random seeds, stratified splitting, and SMOTE applied only to training data; reliability is ensured through full reproducibility via pinned library versions and version-controlled code.

3.4 Explainability Methods

3.4.1 SHAP (SHapley Additive exPlanations)

SHAP assigns each feature a contribution score derived from Shapley values in cooperative game theory (Lundberg and Lee, 2017). The method satisfies three axiomatic properties - local accuracy, missingness, and consistency guaranteeing that feature attributions are mathematically well-founded. For any individual customer, the sum of all SHAP values equals the difference between the prediction and the baseline expected value. TreeSHAP (Lundberg, Erion and Lee, 2018) will be used for Random Forest and XGBoost; KernelSHAP for Logistic Regression.

3.4.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME explains individual predictions by constructing a local surrogate model around a single instance (Ribeiro, Singh and Guestrin, 2016). It generates perturbed copies of the input, weights them by proximity, observes the black-box model's predictions, and fits a sparse linear model to approximate the local decision boundary. Implemented using the Python lime package. Because LIME's outputs are sensitive to hyperparameter choices, particularly perturbation samples and kernel width all settings will be held constant across experiments to ensure observed differences reflect method behaviour, not configuration variation.

3.5 Quantitative Evaluation of Explanations

3.5.1 Fidelity

Fidelity measures how accurately an explanation reflects the true behaviour of the underlying model. Operationalised using a remove-and-retrain protocol: top-k features identified by each method are sequentially removed, and the resulting performance drop measured. Greater degradation indicates higher fidelity (Hooker et al., 2019).

3.5.2 Stability

For each test customer, ten small perturbations will be generated by adding minor noise to continuous features. Explanations will be computed for all perturbations and their variance measured. SHAP is expected to demonstrate higher stability than LIME (Krishna et al., 2022; Zhang et al., 2019).

3.5.3 Sparsity

Measured by counting features whose absolute attribution exceeds 5% of the maximum attribution for that instance. Marketer-actionable explanations should ideally highlight approximately five to seven features.

3.5.4 Inter-method Agreement

Following Krishna et al. (2022): feature agreement (overlap in top 5 feature sets), rank agreement (correlation of feature rankings), and sign agreement (directional consistency), aggregated across the test set.

3.6 SHAP-Based Churn Segmentation

The segmentation component represents the novel practical contribution. The goal is not simply to predict that a customer will churn, but to identify why, whether the driver is price sensitivity, service dissatisfaction, competitor appeal, or feature gaps so that the business can respond with the right intervention for each group. Predicted churners are isolated, the SHAP feature-value vector for each is extracted, and k-means clustering is applied to group churners into reason-based segments. Each cluster is described by its dominant SHAP values . For instance, a "price-sensitive" segment dominated by monthly charges and contract length is targeted with a discount or plan downgrade offer; a "service-dissatisfied" segment dominated by customer service call frequency triggers proactive outreach and service recovery. Each segment is mapped to a specific, actionable retention strategy that a marketing team can deploy immediately.

3.7 Streamlit Application

An interactive Streamlit dashboard will make the framework accessible to non-technical users. Marketers will be able to search or filter customers, view churn probability with a confidence indicator, toggle between SHAP and LIME explanation visualisations side-by-side, explore the customer's assigned churn segment with its reason-profile, and receive an automated retention recommendation tailored to that segment. The interface will include interactive charts, dropdown filters, and a segment comparison view allowing marketers to compare reason-profiles across clusters. Hosted on GitHub with a pinned requirements.txt for full reproducibility.

3.8 Ethical Considerations

The Cell2Cell dataset is fully anonymised and publicly available; no fresh ethics approval is required. This will be confirmed with Carolyn Barker. Bias and fairness audits will be conducted across age and gender using demographic parity and equalised odds (Selbst and Barocas, 2018). The risk that explanation methods can be manipulated to mask discriminatory behaviour is acknowledged (Wachter, Mittelstadt and Russell, 2018). These risks are addressed in future work.

3.9 Tools, Environment, and Reproducibility

All experiments will be implemented in Python 3.11+ using the following core libraries: scikit-learn for classification and clustering, XGBoost for gradient boosting, imbalanced-learn for SMOTE, SHAP and LIME for explanation generation, and Streamlit for the demonstration tool. Visualisation will use matplotlib, seaborn, and plotly. Development will be done on a Windows environment using VS Code and Jupyter notebooks. All code will be controlled with Git and will be hosted on GitHub. Random seeds will be fixed at 42 across every probabilistic operation, and all library versions will be pinned in "requirements.txt" to guarantee full reproducibility in the GitHub repository.

4. Timetable and Risk Management

4.1 Project Timeline

The project spans sixteen weeks from 5 May to 24 August 2026, structured across fourteen phases balancing sequential dependencies with deliberate parallelism. The timetable includes 3 milestones: submission of AE1 (15 June), submission of AE2 (21 August) and presentation of AE3 (week beginning 24 August).

- Phases 1-3 (Weeks 1-5): supervisor onboarding, literature review (strict internal deadline of 8 June), proposal writing
- The analytical core after AE1: Phase 4 (Weeks 6-7) - Data collecting, EDA and pre-processing.
- Phase 5 (Weeks 7–8) is focused on model training and hyperparameter optimisation.
- Phase 6 (Weeks 8–9) applies SHAP and LIME to the best performing model.
- Phase 7 (Weeks 9-10) Performs the four-metric evaluation.
- Phase 8 (Weeks 10-11) performs SHAP-based clustering.
- Phase 9 (Weeks 11–12) builds the Streamlit tool.
- Critically, dissertation writing (Phase 11) begins in Week 11 and runs in parallel with Phases 8 and 9, avoiding the common pitfall of compressing 10,000 words into the final fortnight.

The target is a complete first draft by 4 August, leaving seventeen days for revision. Approximately ten days of slack are distributed across Phases 4–5 and 10–11 to absorb delays.

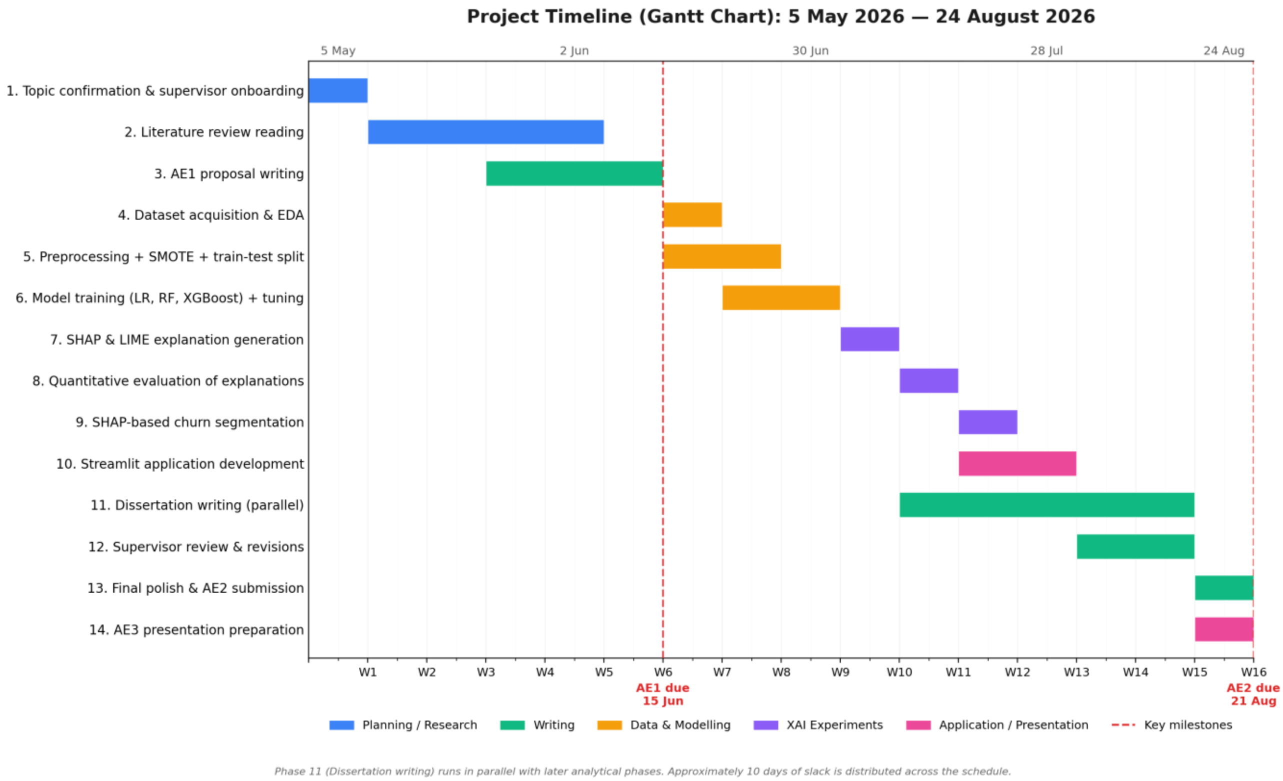

Figure 1. Project Gantt chart, 5 May 2026 to 24 August 2026.

4.2 Risk Register

Ten risks have been identified, assessed by likelihood and impact, and paired with specific mitigation strategies. Table 1 presents the full register.

Risk	Likelihood	Impact	Mitigation
Dataset quality issues (missing values, label leakage, encoding inconsistencies)	Medium	High	3-day buffer in Phase 4; IBM Telco (7,043 rows) as fallback dataset; full EDA before modelling.
XGBoost training/tuning exceeds laptop capacity on 71k rows	Medium	Medium	Google Colab as compute fallback; reduce grid search resolution if needed.
LIME produces highly unstable explanations	High	Low	Expected finding; document and characterise the variance. Fixed seeds and multiple sampling runs.
SHAP and LIME fundamentally disagree with no clear winner	High	Low	Valid research finding. Reinforces the disagreement-problem literature.
SHAP-vector clusters lack clear interpretable reason-profiles	Medium	High	Try multiple clustering methods (k-means, hierarchical, DBSCAN); apply PCA before clustering; report as a finding about feature-attribution limits if persistent.
Literature review takes longer than planned	High	High	Hard deadline on lit review (8 June); accept good-enough over perfect; biweekly supervisor check-ins.
Supervisor unavailable during critical review periods	Medium	Medium	Confirm availability at first meeting; book sessions in advance; agree on async feedback workflow.
Scope creep - adding counterfactuals, deep learning, NLP on feedback text	High	Medium	Explicit scope statement in proposal; relegate additional methods to Future Work; supervisor enforces scope discipline.
Reproducibility failure: code runs differently each execution	Medium	Low	Fixed seeds throughout; requirements.txt pinned; weekly commits to GitHub.
AE2 deadline pressure: 10,000 words in final two weeks	Medium	High	Write in parallel with later analytical phases (Phase 11 starts in W11); first complete draft by 4 August, leaving 17 days for revision.

Table 1. Risk register with likelihood, impact, and mitigation.

The timetable includes built-in slack distributed across the schedule, and the risk register will be revisited at each supervisor meeting to ensure emerging risks are captured early and mitigations remain current.

References

Adadi, A. and Berrada, M. (2018) 'Peeking inside the black-box: a survey on explainable artificial intelligence (XAI)', *IEEE Access*, 6, pp. 52138–52160. Available at: https://doi.org/10.1109/ACCESS.2018.2870052 (Accessed: 12 June 2026).

Al Mamun, A., Tanoor, M., Masum, A., Bhuiyan, T. and Hassan, M. (2026) 'Explainable AI for customer churn prediction in the energy sector using ensemble machine learning models', *EAI Endorsed Transactions on AI and Robotics*, 5. Available at: https://publications.eai.eu/index.php/airo/article/view/9813 (Accessed: 12 June 2026).

Bhatt, U., Xiang, A., Sharma, S., Weller, A., Taly, A., Jia, Y., Ghosh, J., Puri, R., Moura, J. and Eckersley, P. (2020) 'Explainable machine learning in deployment', in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. Available at: https://arxiv.org/abs/1909.06342 (Accessed: 12 June 2026).

Bolla, S. and Narsepalle, K. (2026) 'Enhancing customer churn prediction in telecommunications with explainable AI: a SHAP-based approach', in *Computing and Machine Learning: CML 2025*. Lecture Notes in Networks and Systems, vol. 1613. Singapore: Springer. Available at: https://doi.org/10.1007/978-981-95-2875-2_1 (Accessed: 7 June 2026).

Breiman, L. (2001) 'Random forests', *Machine Learning*, 45(1), pp. 5–32. Available at: https://doi.org/10.1023/A:1010933404324 (Accessed: 12 June 2026).

Chen, T. and Guestrin, C. (2016) 'XGBoost: a scalable tree boosting system', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. Available at: https://arxiv.org/abs/1603.02754 (Accessed: 12 June 2026).

Doshi-Velez, F. and Kim, B. (2017) 'Towards a rigorous science of interpretable machine learning', *arXiv preprint*. Available at: https://arxiv.org/abs/1702.08608 (Accessed: 12 June 2026).

El Attar, A. and El-Hajj, M. (2026) 'Explainable AI-driven customer churn prediction: a multi-model ensemble approach with SHAP-based feature analysis', *Frontiers in Artificial Intelligence*, 8. Available at: https://doi.org/10.3389/frai.2026.1748799 (Accessed: 12 June 2026).

Explainable customer churn prediction in telecom using ensemble learning and SHAP analysis (2024) *Mathematical Modelling of Engineering Problems*, 12(12). Available at: https://www.iieta.org/journals/mmep/paper/10.18280/mmep.121226 (Accessed: 12 June 2026).

Hooker, S., Erion, G., Kim, B. and Lee, S. (2019) 'A benchmark for interpretability methods in deep neural networks', *Advances in Neural Information Processing Systems*, 32. Available at: https://arxiv.org/abs/1806.10758 (Accessed: 12 June 2026).

Imani, M., Beikmohammadi, A. and Arabnia, H. (2025) 'Comprehensive analysis of Random Forest and XGBoost performance with SMOTE, ADASYN, and GNUS under varying imbalance levels', *Technologies*, 13(3), 88. Available at: https://www.mdpi.com/2227-7080/13/3/88 (Accessed: 12 June 2026).

Krishna, S., Han, T., Gu, A., Pombra, J., Jabbari, S., Wu, S. and Lakkaraju, H. (2022) 'The disagreement problem in explainable machine learning: a practitioner's perspective', *Transactions on Machine Learning Research*. Available at: https://arxiv.org/abs/2202.01602 (Accessed: 12 June 2026).

Lundberg, S. and Lee, S. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30. Available at: https://arxiv.org/abs/1705.07874 (Accessed: 12 June 2026).

Lundberg, S., Erion, G. and Lee, S. (2018) 'Consistent individualized feature attribution for tree ensembles', *arXiv preprint*. Available at: https://arxiv.org/abs/1802.03888 (Accessed: 12 June 2026).

Molnar, C. (2025) *Interpretable machine learning: a guide for making black box models explainable*. 3rd edn. Available at: https://christophm.github.io/interpretable-ml-book/ (Accessed: 12 June 2026).

Nurhidayat, M. and Anggraini, D. (2025) 'Evaluation of telecommunication customer churn classification with SMOTE using Random Forest and XGBoost algorithms', *Journal of Applied Informatics and Computing*, 9(1), pp. 89–95. Available at: https://jurnal.polibatam.ac.id/index.php/JAIC/article/view/8740 (Accessed: 12 June 2026).

Pandey, P. and Pandey, S. (2026) 'A systematic literature review on explainable AI for churn prediction, customer segmentation and retention', *International Journal of Data Science and Analytics*. Available at: https://doi.org/10.1007/s41060-025-01018-0 (Accessed: 12 June 2026).

Ribeiro, M., Singh, S. and Guestrin, C. (2016) '"Why should I trust you?": explaining the predictions of any classifier', in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. New York: ACM, pp. 1135–1144. Available at: https://arxiv.org/abs/1602.04938 (Accessed: 12 June 2026).

Rudin, C. (2019) 'Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead', *Nature Machine Intelligence*, 1(5), pp. 206–215. Available at: https://arxiv.org/abs/1811.10154 (Accessed: 12 June 2026).

Saunders, M., Lewis, P. and Thornhill, A. (2019) *Research methods for business students*. 8th edn. Harlow: Pearson Education.

Selbst, A. and Barocas, S. (2018) 'The intuitive appeal of explainable machines', *Fordham Law Review*, 87, pp. 1085–1139. Available at: https://ir.lawnet.fordham.edu/flr/vol87/iss3/11/ (Accessed: 12 June 2026).

Slack, D., Hilgard, S., Jia, E., Singh, S. and Lakkaraju, H. (2020) 'How can we fool LIME and SHAP? Adversarial attacks on post hoc explanation methods', in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*. Available at: https://arxiv.org/abs/1911.02508 (Accessed: 12 June 2026).

Verbeke, W., Dejaeger, K., Martens, D., Hur, J. and Baesens, B. (2012) 'New insights into churn prediction in the telecommunication sector: a profit driven data mining approach', *European Journal of Operational Research*, 218(1), pp. 211–229. Available at: https://www.researchgate.net/publication/220288606 (Accessed: 12 June 2026).

Wachter, S., Mittelstadt, B. and Russell, C. (2018) 'Counterfactual explanations without opening the black box: automated decisions and the GDPR', *Harvard Journal of Law and Technology*, 31(2), pp. 841–887. Available at: https://arxiv.org/abs/1711.00399 (Accessed: 12 June 2026).

Zhang, Y., Song, K., Sun, Y., Tan, S. and Udell, M. (2019) 'Why should you trust my explanation? Understanding uncertainty in LIME explanations', *arXiv preprint*. Available at: https://arxiv.org/abs/1904.12991 (Accessed: 12 June 2026).